Structured notes: Abstract Algebra I, Module 3

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Module 3 (sections 4.8–4.15, 8.19–8.25, 10.1–10.20). The syllabus names the 7th edition; items are matched by content. Numbers are the book's own. Formal version of the same content, with numbered results and the same order of items. Exercises: only the exercises the text names are reproduced, each directly after the item that names it. Figures are described with their labels; not redrawn. Items 4.1–4.7 and 4.16 onward, Section 9 and Section 11 are not in this module.

Section 4: Permutations and Disjoint Cycles (4.8–4.15)

Printed pp. 43–46 (4.8–4.15; Table 4.15)

Prerequisites

- Group axioms (Definition 2.1); isomorphisms (Section 2).
- Composition and inverses of bijections.
- Two-row notation for permutations (Section 4, printed 39–42, not reproduced).

1. The group of permutations

THEOREM 4.8 Let A be a nonempty set and S_A the set of all permutations of A. Then S_A is a group under permutation multiplication.

Proof. Composition of permutations is a permutation, so S_A is closed. Composition is associative, so $\mathcal{G}_1$ holds. The identity map ι satisfies $\mathcal{G}_2$. A permutation σ is one-to-one and onto, so it has an inverse function σ^{-1}, a permutation with $\sigma^{-1}\sigma = \sigma\sigma^{-1} = \iota$; this gives $\mathcal{G}_3$. ◆

Warning. Some texts multiply permutations left to right, so that $(\sigma\mu)(a) = \mu(\sigma(a))$. Exercise 34 asks the reader to check that both conventions give a group. The text of Exercise 34 follows.

Exercise 34. (See the warning following Theorem 4.8.) Let G be a group with binary operation $*$. Let G' be the same set as G, and define a binary operation $*'$ on G' by $x *' y = y * x$ for all $x, y \in G'$. (a) Intuitive argument that G' under $*'$ is a group: suppose the front wall of your classroom were made of transparent glass, and that all possible products $a * b = c$ and all possible instances $a * (b * c) = (a * b) * c$ of the associative property for G under $*$ were written on the wall with a magic marker. What would a person see when looking at the other side of the wall from the next room in front of yours? (b) Show from the mathematical definition of $*'$ that G' is a group under $*'$.

If $|A| = |B|$, then $S_A \simeq S_B$. A one-to-one map $f : A \to B$ onto gives the isomorphism $\phi(\sigma) = \bar{\sigma}$, where $\bar{\sigma}(f(a)) = f(\sigma(a))$. So the structure of S_A depends only on $|A|$, and $\{1, \ldots, n\}$ is a prototype for any set of n elements. The book's example: $A = \{1, 2, 3\}$, $B = \{\#, \$, \%\}$, $f(1) = \#$, $f(2) = \$$, $f(3) = \%$ sends $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$ to $\begin{pmatrix} \# & \$ & \% \\ \% & \$ & \# \end{pmatrix}$.

DEFINITION 4.9 For $A = \{1, \ldots, n\}$, the group of all permutations of A is the **symmetric group on n letters**, S_n. It has $n!$ elements.

Remark (not numbered). S_3 is not abelian: with $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$, $\sigma\tau(1) = 2 \neq 3 = \tau\sigma(1)$. Since every group of order at most 4 is abelian and $\mathbb{Z}_5$ is the only group of order 5, S_3 is the smallest nonabelian group.

PLUS TWO RECALL: INVERSE FUNCTIONS (STATUS OPEN, CHECK AGAINST THE KERALA TEXT)

Recall. A function is one-to-one and onto exactly when it has an inverse function. The inverse reverses each arrow: if $\sigma(a) = b$ then $\sigma^{-1}(b) = a$.

Re-derive.

1. For $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$: $\sigma(1) = 3$, so $\sigma^{-1}(3) = 1$; reading the columns backward gives $\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$.
2. Composing: $\sigma^{-1}(\sigma(a)) = a$ for every a.

Saved. Example 4.10 (inverting a permutation by swapping the rows).

Example 4.10. The inverse of $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 6 & 1 & 4 & 5 \end{pmatrix}$ is $\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 2 & 1 & 5 & 6 & 3 \end{pmatrix}$, obtained by interchanging the rows and sorting the columns.

2. Disjoint cycles

For $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 6 & 2 & 5 & 1 \end{pmatrix}$ we write $(1, 3, 6)(2, 4)$: the cycle $(1, 3, 6)$ is a 3-cycle, and $(2, 4)$

is a 2-cycle. Entries that are fixed are omitted. A 2-cycle is a **transposition**. Cycles are **disjoint** if no entry lies in two of them. Every permutation of S_n has a disjoint cycle form, unique up to order and cyclic rotation.

Example 4.11.
$(1,5,2,7)(3,4,9) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 7 & 4 & 9 & 2 & 6 & 1 & 8 & 3 \end{pmatrix}$
.

The operation is composition, so the rightmost cycle acts first.

Example 4.12. In S_6,
$(1,5,3,2,6)(1,2,4,3,6) = (1,6,5,3)(2,4)$.

Example 4.13.
$(1,5)(2,4)(1,4,3)(2,5)(4,2,1) = (1,3,5,4)$. The calculation does not use the value of n and holds in S_n for every $n \geq 5$.

Example 4.14. $(1,5,7)(3,8,2,4,6)$ has inverse $(6,4,2,8,3)(7,5,1)$, since $(ab)^{-1} = b^{-1}a^{-1}$ and the inverse of a cycle is the cycle backward; this also equals $(1,7,5)(2,8,3,6,4)$.

4.15 Table The multiplication table of S_3. Entry (row x, column y) is $x \circ y$.

$S_3, \circ$	ι	$(1,2,3)$	$(1,3,2)$	$(1,2)$	$(1,3)$	$(2,3)$
ι	ι	$(1,2,3)$	$(1,3,2)$	$(1,2)$	$(1,3)$	$(2,3)$
$(1,2,3)$	$(1,2,3)$	$(1,3,2)$	ι	$(1,3)$	$(2,3)$	$(1,2)$
$(1,3,2)$	$(1,3,2)$	ι	$(1,2,3)$	$(2,3)$	$(1,2)$	$(1,3)$
$(1,2)$	$(1,2)$	$(2,3)$	$(1,3)$	ι	$(1,3,2)$	$(1,2,3)$
$(1,3)$	$(1,3)$	$(1,2)$	$(2,3)$	$(1,2,3)$	ι	$(1,3,2)$
$(2,3)$	$(2,3)$	$(1,3)$	$(1,2)$	$(1,3,2)$	$(1,2,3)$	ι

The table is not symmetric about the diagonal, so S_3 is nonabelian. Disjoint cycles commute;
$(1,2)(2,3) = (1,2,3) \neq (1,3,2) = (2,3)(1,2)$.

Items 4.16 onward (dihedral groups) are outside these notes.

Section 8: Even and Odd Permutations, Alternating Groups (8.19–8.25)

Printed pp. 82–84 (8.19–8.25; Figures 8.20, 8.21)

Prerequisites

- Products of transpositions (8.14–8.15); orbits (8.17–8.18), Module 2.
- Kernels and homomorphisms (8.5–8.6), Module 2.

1. Parity of permutations

THEOREM 8.19 No permutation in S_n is both a product of an even number and a product of an odd number of transpositions.

Proof. Let $\sigma \in S_n$ and $\tau = (i, j)$. Claim: the number of orbits of $\tau\sigma$ and of σ differ by 1.

Case I (i, j in different orbits of σ). Write σ with cycles $(b, j, \times, \times, \times)(a, i, \times, \times)$ (Figure 8.20). Then $(i, j)(b, j, \times, \times, \times)(a, i, \times, \times) = (a, j, \times, \times, \times, b, i, \times, \times)$. Two orbits are joined into one.

Exercise 42. (named in Case I, Theorem 8.19) a. Draw a figure like Fig. 8.20 to illustrate that if i and j are in different orbits of σ and $\sigma(i) = i$, then the number of orbits of $(i, j)\sigma$ is one less than the number of orbits of σ.
b. Repeat part (a) if $\sigma(j) = j$ also.

Case II (i, j in the same orbit). Write σ with first cycle $(a, i, \times, \times, \times, b, j, \times, \times)$ (Figure 8.21). Then $(i, j)(a, i, \times, \times, \times, b, j, \times, \times) = (a, j, \times, \times, \times)(b, i, \times, \times, \times)$. One orbit is split into two.

In both cases the orbit count changes by 1. The identity has n orbits, so the number of orbits of σ differs from n by an even number or by an odd number, not both. Hence σ cannot be written with an even number of transpositions and also with an odd number. ◆

Figures 8.20 and 8.21 show the two cases as circles joined by a pinched middle (8.20) and as one loop split into two (8.21), with the points labelled i, j, a, b. Labels as printed; not redrawn.

DEFINITION 8.22 A permutation of a finite set is **even** (respectively **odd**) if it is a product of an even (respectively odd) number of transpositions.

Example 8.23. $\iota = (1, 2)(1, 2)$ is even (and is defined to be even for $n = 1$).
$(1, 4, 5, 6)(2, 1, 5) = (1, 6)(1, 5)(1, 4)(2, 5)(2, 1)$ has five transpositions, so it is odd.

2. The alternating groups

For $n \geq 2$ the sets A_n (even) and B_n (odd) have equal size. With $\tau = (1, 2)$, the map $\lambda_\tau(\sigma) = \tau\sigma$ takes A_n into B_n, is one-to-one by cancellation, and is onto since $\lambda_\tau(\tau^{-1}\rho) = \rho$ for $\rho \in B_n$.

The product of two even permutations is even, ι is even, and σ^{-1} is even when σ is (reverse the transpositions). So the even permutations form a subgroup.

THEOREM 8.24 If $n \geq 2$, the even permutations of $\{1, \ldots, n\}$ form a subgroup of S_n of order $n!/2$.

Sign. The book uses $\mathbf{sgn}$ in the standard definition of the determinant and asks for the facts below (Exercise 52, printed next). $\mathbf{sgn} : S_n \to \{1, -1\}$, $\mathbf{sgn}(\sigma) = 1$ if σ is even and -1 if odd, is a homomorphism; $\mathbf{Ker}(\mathbf{sgn})$ is the set of even permutations. It defines the determinant.

Exercise 52. (named after Theorem 8.24, in the sign paragraph) The usual definition for the determinant of an $n \times n$ matrix $A = (a_{ij})$ is

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{1,\sigma(1)} a_{2,\sigma(2)} a_{3,\sigma(3)} \cdots a_{n,\sigma(n)},$$

where $\operatorname{sgn}(\sigma)$ is the sign of σ. Using this definition, prove the following properties of determinants.
a. If a row of matrix A has all zero entries, then $\det(A) = 0$.
b. If two different rows of A are switched to obtain B, then $\det(B) = -\det(A)$.
c. If r times one row of A is added to another row of A to obtain a matrix B, then $\det(A) = \det(B)$.
d. If a row of A is multiplied by r to obtain the matrix B, then $\det(B) = r \det(A)$.

DEFINITION 8.25 The subgroup of S_n of even permutations of n letters is the **alternating group** A_n on n letters.

Cayley's theorem shows that every finite group of order n is a subgroup of S_n up to isomorphism. The absence of radical formulas for polynomial equations of degree $n \geq 5$ is due to the structure of A_n.

Section 10: Cosets and the Theorem of Lagrange (10.1–10.20)

Printed pp. 97–104 (10.1–10.20; Tables 10.14, 10.15; Example 10.16)

Prerequisites

- Subgroup tests (Theorems 5.12 and 5.16); equivalence relations and their classes (Section 0).
- Homomorphisms and kernels (Section 8, Definition 8.6).

1. Cosets

THEOREM 10.1 Let $H \le G$ and define $a \sim_L b$ if and only if $a^{-1}b \in H$. Then $\sim_L$ is an equivalence relation on G.

Proof. Reflexive: $a^{-1}a = e \in H$. Symmetric: $(a^{-1}b)^{-1} = b^{-1}a \in H$. Transitive: $(a^{-1}b)(b^{-1}c) = a^{-1}c \in H$. ◆

The class of a is $\{x \mid a^{-1}x \in H\} = \{ah \mid h \in H\}$.

DEFINITION 10.2 For $H \le G$ and $a \in G$, $aH = \{ah \mid h \in H\}$ is the **left coset** of H containing a.

Example 10.3. The left cosets of $3\mathbb{Z}$ in $\mathbb{Z}$ are $3\mathbb{Z}$, $1 + 3\mathbb{Z}$ and $2 + 3\mathbb{Z}$.

Example 10.4. The left cosets of $\langle 3 \rangle$ in $\mathbb{Z}_{12}$ are $\{0,3,6,9\}$, $\{1,4,7,10\}$, $\{2,5,8,11\}$.

Example 10.5. The left cosets of $\langle \mu \rangle = \{\iota, \mu\}$ in D_4: $\{\iota, \mu\}$, $\{\rho, \mu\rho^3\}$, $\{\rho^2, \mu\rho^2\}$, $\{\rho^3, \mu\rho\}$.

2. The theorem of Lagrange

THEOREM 10.6 For $H \le G$ and $a \in G$, aH has the same cardinality as H.

Proof. $f(h) = ah$ is one-to-one by left cancellation, and onto aH by definition. ◆

THEOREM 10.7 (THEOREM OF LAGRANGE) If $H \le G$ and G is finite, then $|H|$ divides $|G|$.

Proof. Each coset has $|H|$ elements and the cosets partition G; if there are r of them, $|G| = r|H|$. ◆

COROLLARY 10.8 Every group of prime order is cyclic.

Proof. A generator $a \neq e$ gives $|\langle a \rangle|$ dividing the prime p and at least 2; so $\langle a \rangle = G$. ◆

There is therefore one group of each prime order, up to isomorphism.

THEOREM 10.9 The order of an element of a finite group divides the order of the group.

Proof. The order of a is $|\langle a \rangle|$; apply Theorem 10.7. ◆

DEFINITION 10.10 The number of left cosets of H in G is the **index** $(G : H)$.

If G is finite, $(G : H) = |G|/|H|$. The proof of Theorem 10.11 is left to the exercises, which the text names as Exercise 40, printed next.

Exercise 40. Prove Theorem 10.11 [*Hint:* Let $\{a_iH \mid i = 1, \ldots, r\}$ be the collection of distinct left cosets of H in G and $\{b_jK \mid j = 1, \ldots, s\}$ be the collection of distinct left cosets of K in H. Show that $\{(a_ib_j)K \mid i = 1, \ldots, r;\ j = 1, \ldots, s\}$ is the collection of distinct left cosets of K in G.]

THEOREM 10.11 If $K \le H \le G$ and $(H : K)$, $(G : H)$ are finite, then $(G : K)$ is finite and $(G : K) = (G : H)(H : K)$.

The converse of Lagrange's theorem is false in general; Section 13 of the book gives the example. For abelian groups the answer is in Section 9.

Right cosets

With $a \sim_R b$ if and only if $ab^{-1} \in H$, the classes are the right cosets $Ha = \{ha \mid h \in H\}$. Each has the cardinality of H. In abelian groups left and right cosets coincide; in general they need not.

Example 10.12. The right cosets of $\{\iota, \mu\}$ in D_4 are $\{\iota, \mu\}$, $\{\rho, \mu\rho\}$, $\{\rho^2, \mu\rho^2\}$, $\{\rho^3, \mu\rho^3\}$. Since $\rho H = \{\rho, \mu\rho^3\} \neq \{\rho, \mu\rho\} = H\rho$, they differ from the left cosets.

Example 10.13. In $\mathbb{Z}_6$ with $H = \{0, 3\}$ the cosets are $\{0,3\}$, $\{1,4\}$, $\{2,5\}$.

10.14 Table The addition table of $\mathbb{Z}_6$ with elements in the order of the cosets. Shading as printed: 0 and 3 unshaded, 1 and 4 light, 2 and 5 dark.

$+_6$	**0**	**3**	**1**	**4**	**2**	**5**
0	0	3	1	4	2	5
3	3	0	4	1	5	2
1	1	4	2	5	3	0
4	4	1	5	2	0	3
2	2	5	3	0	4	1
5	5	2	0	3	1	4

10.15 Table The operation on the cosets LT, MD, DK.

	LT	**MD**	**DK**
LT	LT	MD	DK
MD	MD	DK	LT
DK	DK	LT	MD

With LT $= 0$, MD $= 1$, DK $= 2$ this is the table of $\mathbb{Z}_3$. The construction works in general only when left and right cosets are the same (Section 12).

Example 10.16. The group table of D_3 with the left cosets of $\{\iota, \mu\}$ grouped: rows and columns $\iota, \mu \mid \rho, \mu\rho^2 \mid \rho^2, \mu\rho$. Most blocks contain elements of more than one coset, and the second row of blocks contains two blocks with the same elements $\{\rho^2, \mu\rho, \mu, \iota\}$; so there is no natural group of blocks.

	ι	μ	ρ	$\mu\rho^2$	ρ^2	$\mu\rho$
ι	ι	μ	ρ	$\mu\rho^2$	ρ^2	$\mu\rho$
μ	μ	ι	$\mu\rho$	ρ^2	$\mu\rho^2$	ρ
ρ	ρ	$\mu\rho^2$	ρ^2	$\mu\rho$	ι	μ
$\mu\rho^2$	$\mu\rho^2$	ρ	μ	ι	$\mu\rho$	ρ^2
ρ^2	ρ^2	$\mu\rho$	ι	μ	ρ	$\mu\rho^2$
$\mu\rho$	$\mu\rho$	ρ^2	$\mu\rho^2$	ρ	μ	ι

3. Kernels and cosets

THEOREM 10.17 For a homomorphism $\phi : G \to G'$, the left and right cosets of $\mathbf{Ker}(\phi)$ are identical; and a, b lie in the same coset of $\mathbf{Ker}(\phi)$ if and only if $\phi(a) = \phi(b)$.

Proof. If $a^{-1}b \in \mathbf{Ker}(\phi)$ then $\phi(a) = \phi(b)$, and $\phi(ab^{-1}) = e$, so $ab^{-1} \in \mathbf{Ker}(\phi)$. If $\phi(a) = \phi(b)$ then $b^{-1}a \in \mathbf{Ker}(\phi)$. The right-coset case is left to the reader. ◆

Example 10.18. For $\det : GL(2, \mathbb{R}) \to \mathbb{R}^*$, two matrices lie in the same coset of the kernel if and only if they have the same determinant. Both $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$ have determinant 2.

COROLLARY 10.19 A homomorphism $\phi : G \to G'$ is one-to-one if and only if $\mathbf{Ker}(\phi)$ is trivial.

Proof. Apply Theorem 10.17. ◆

Example 10.20. $\phi : \mathbb{R} \to \mathbb{R}^+$, $\phi(x) = 2^x$, is one-to-one, since $2^x = 1$ only for $x = 0$.

Notes on this section

This section covers book items 10.1 to 10.20 and Tables 10.14, 10.15 and the D_3 table of Example 10.16. Items 10.1 to 10.9 are syllabus item 15 and 10.10 to 10.14 are item 16; 10.15 to 10.20 complete the subsection. Exercise 40 is printed after the sentence that names it.